

BASIC PRINCIPLES OF MEDICINE 1: **FOUNDATIONS TO MEDICINE**

Transcription

Factors That Disrupt Transcription

Cristofre Martin
cmartin@sgu.edu

Objectives:

SOM.MK.I.BPM1.1.FTM.3.BCHM.0057	Explain the mechanism of action of rifampin and alpha-amanitin as inhibitors of transcription.
SOM.MK.I.BPM1.1.FTM.3.BCHM.0058	Describe the effect of the antibiotic drugs that interfere with DNA gyrase such as (quinolones) ciprofloxacin

The mushroom *Amanita phalloides* (The Death Cap Mushroom) produces a toxin called alpha-amanitin which is a potent inhibitor of RNA polymerase II.

*** The microfilament toxin, Phalloidin, that is also found in this mushroom is often erroneously cited for its toxicity. Phalloidin is not absorbed by the GI tract and does not contribute to toxicity by oral intake.

There are four clinical phases of poisoning:

- 1) Asymptomatic phase**
- 2) Gastrointestinal phase (1-2 days) - severe diarrhea and vomiting**
- 3) Apparent recovery phase - few symptoms**
- 4) Hepatic phase - renal and liver failure**

*** death may occur in 7-10 days

Treatment:

- gastrointestinal decontamination
- high dose penicillin (inhibits amanitin activity in liver)
- liver transplant



Rifampin (Rifampicin)

-an antibiotic from a strain of Streptomyces

-inhibits initiation of transcription

-blocks formation of the first phosphodiester bond by binding to prokaryote RNA polymerase

-RNA polymerase of eukaryotes is not affected

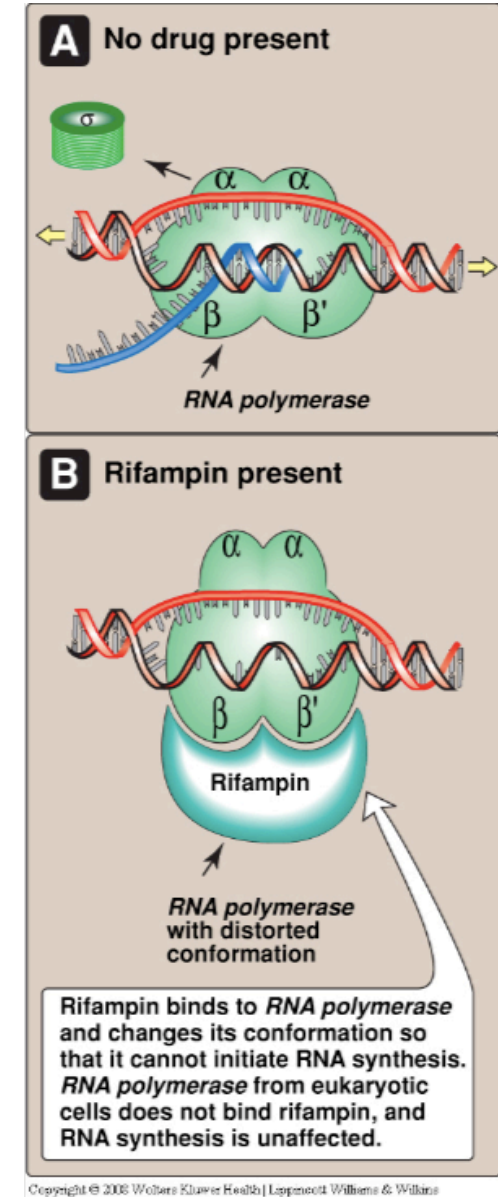
Used in the treatment of:

Tuberculosis

Leprosy

Mycobacterium infections

*** Mutations in the genes encoding the target proteins such as RNA polymerase and topoisomerases can lead to drug resistance.

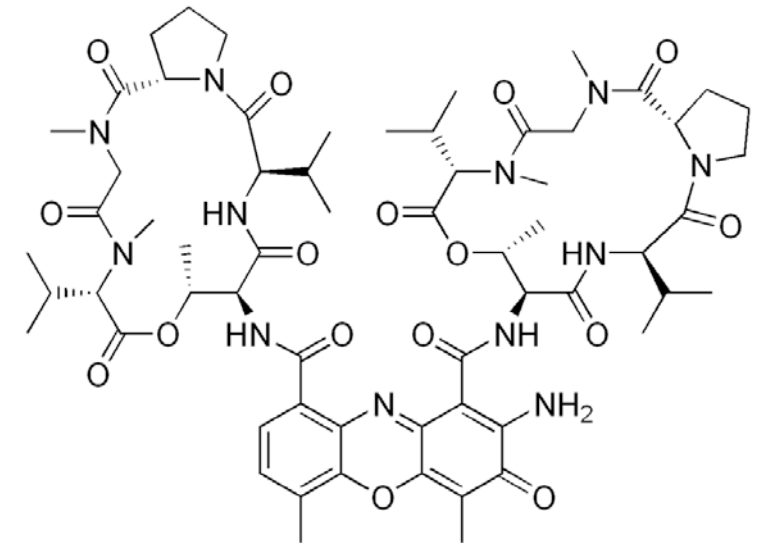


Intercalating Inhibitors of RNA synthesis:

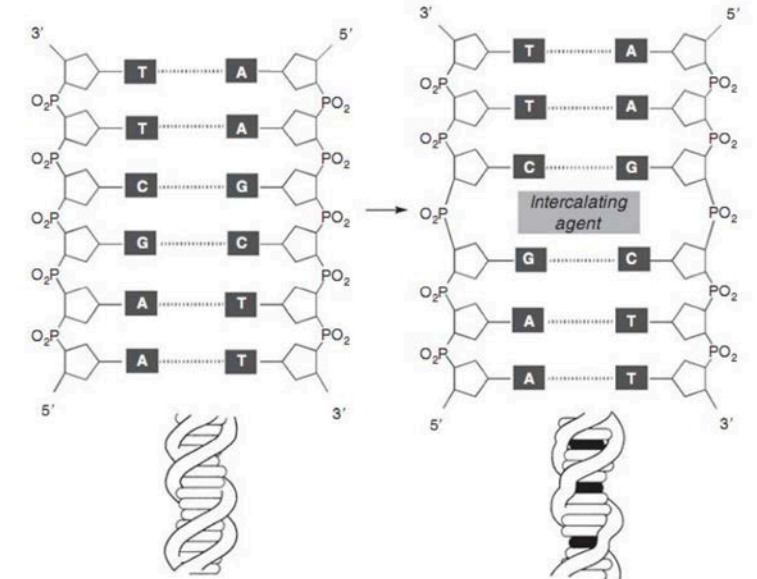
Actinomycin D

- binds tightly to double stranded DNA
- inhibits transcription and replication in both prokaryotes and eukaryotes
- not normally used as an antibiotic

Used as an anti-cancer drug (inhibiting DNA replication and cell division) for the treatment of Wilms tumor, rhabdomyosarcoma, Ewing's sarcoma, trophoblastic neoplasm, testicular cancer, and certain types of ovarian cancer



Actinomycin D



Transcription results in positive (+ve) supercoils in front of the RNA polymerase and negative (-ve) supercoils behind the RNA polymerase.

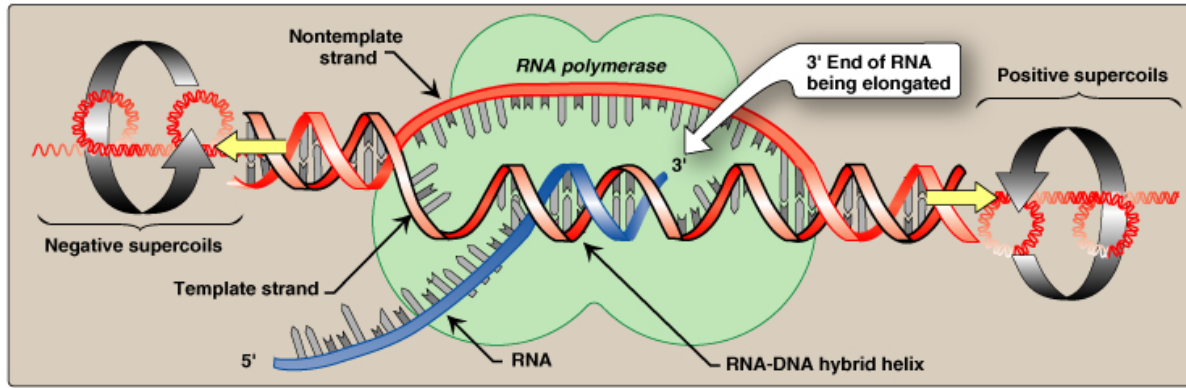
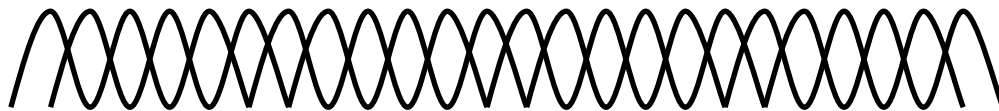


Figure 30.8
Local unwinding of DNA caused by *RNA polymerase*.

Topoisomerase
Relaxes -ve supercoils

Gyrase (another topoisomerase)
Introduces -ve supercoils



During transcription (and DNA replication), supercoiling is generated as a side effect of the unwinding of DNA by either helicase (replication) and RNA polymerase (transcription). **If this is not corrected by gyrase (prokaryotes) or topoisomerase (eukaryotes) these processes will halt.**

Bacterial gyrase (topoisomerases) can be inhibited by the antibiotics such as:

Fluoroquinolones (eg. Ciprofloxacin)

Used to treat: infections of bones and joints, endocarditis, gastroenteritis, respiratory tract infections, urinary tract infection and anthrax amongst others.